CSC 481.1: Object-Oriented Programming with Java: Lecture Schedule

Course Description:
This 13-week course covers the principles of Object-Oriented Programming (OOP) using Java. Students will learn the fundamentals of OOP, including classes, objects, inheritance, polymorphism, and encapsulation. They will also gain hands-on experience with Java programming, including data types, control structures, methods, and exception handling.

Course Learning Outcomes:
- Understand the principles of OOP and its applications
- Design and implement Java programs using OOP concepts
- Use Java syntax and data types effectively
- Apply problem-solving skills to real-world problems using Java

Week 1: Introduction to OOP and Java

Learning Outcomes:
- Understand the basics of OOP and its importance
- Familiarize with Java and its features
- Set up Java development environment

Topics:
- Introduction to OOP concepts (classes, objects, inheritance, polymorphism)
- Overview of Java and its features
- Setting up Java development environment
- Basic Java syntax and data types

Week 2: Control Structures, Methods, Classes and Objects

Learning Objectives:
- Apply conditional statements and loops to control program flow.
- Define and invoke methods with parameters and return values.
- Demonstrate method overloading for code flexibility.
- Develop small programs using modular design.
- Define classes and create objects
- Understand constructors and object initialization
- Apply access modifiers (public, private, protected)

Topics:
- Conditionals, loops, methods, method overloading
- Defining classes and creating objects
- Constructors and object initialization
- Access modifiers (public, private, protected)
- Class variables and methods

Week 3: Inheritance

Learning Outcomes:
- Understand inheritance and its types
- Apply single inheritance using the "extends" keyword
- Override methods using method overriding

Topics:
- Introduction to inheritance
- Single inheritance (extends keyword)
- Method overriding
- Super keyword

Week 4: Polymorphism

Learning Outcomes:
- Understand polymorphism and its types

- Apply method overloading and method overriding
- Use dynamic method dispatch

Topics:

- Introduction to polymorphism
- Method overloading
- Method overriding
- Dynamic method dispatch

Week 5: Encapsulation and Abstraction

Learning Outcomes:

- Understand encapsulation and abstraction
- Apply encapsulation using accessor and mutator methods
- Use abstract classes and methods

Topics:

- Encapsulation (hiding implementation details)
- Abstraction (showing only necessary information)
- Accessor and mutator methods
- Abstract classes and methods

Week 6: Interfaces and Abstract Classes

Learning Outcomes:

- Understand interfaces and abstract classes
- Define and implement interfaces
- Compare abstract classes and interfaces

Topics:

- Introduction to interfaces
- Defining and implementing interfaces
- Abstract classes vs. interfaces
- Multiple inheritance using interfaces

Week 7: Exception Handling

Learning Outcomes:

- Understand exception handling and its importance
- Apply try-catch blocks and throw exceptions
- Create custom exceptions

Topics:

- Introduction to exception handling
- Try-catch blocks
- Throwing and catching exceptions
- Custom exceptions

Week 8: Collections Framework

Learning Outcomes:

- Understand Java Collections Framework
- Use Lists, Sets, Maps, and other collections
- Apply iterators and generics

Topics:

- Introduction to Java Collections Framework
- Lists, Sets, Maps, and other collections
- Iterators and generics

Week 9: File Input/Output

Learning Outcomes:

- Understand file input/output in Java
- Read and write text files

- Read and write binary files

Topics:

- Introduction to file input/output
- Reading and writing text files
- Reading and writing binary files
- Serialization and deserialization

Week 10: Multithreading

Learning Outcomes:

- Understand multithreading and its importance
- Create and run threads
- Apply thread synchronization

Topics:

- Introduction to multithreading
- Creating and running threads
- Thread synchronization
- Inter-thread communication

Week 11: Design Patterns

Learning Outcomes:

- Understand design patterns and their importance
- Apply creational, structural, and behavioural patterns
- Use design patterns in Java programming

Topics:

- Introduction to design patterns
- Creational patterns (Singleton, Factory)
- Structural patterns (Adapter, Bridge)
- Behavioural patterns (Observer, Strategy)

Week 12: GUI Programming

Learning Outcomes:

- Understand GUI programming in Java
- Create GUI components using Swing and AWT
- Handle events in GUI programming

Topics:

- Introduction to GUI programming in Java
- Swing and AWT libraries
- Creating GUI components
- Event handling

Week 13: Review and Project Work

Learning Outcomes:

- Review key OOP concepts
- Apply OOP principles in a Java project
- Present and discuss project work

Topics:

- Review of key OOP concepts
- Project work: Design and implement a Java application using OOP principles
- Presentations and feedback

Assessment:

- Quizzes and assignments (10%)
- Project work (20%)
- Final exam (70%)

Resources:

- Java documentation and tutorials
- OOP textbooks and online resources
- Java IDE (Eclipse, NetBeans, or IntelliJ IDEA)